DS-3002 Data Mining

Assignment 1 · Spring 2026

Unveiling Listening Patterns:
Pattern Mining on Real-World Music Streaming Data

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Dataset: HetRec 2011 – Last.fm 2K

Program: BSDS · FAST-NUCES

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Total Marks: 100 · **Weightage:** 5%

Algorithm:	MSApriori
Library:	Python 3.13, pandas, NumPy, matplotlib
Threshold:	Playcount > 200
MIS Scheme:	Tiered
ϕ**:**	0.10
min_conf:	60%

Companion notebook: `i232634_DS-A_Assignment1_Solution.ipynb`

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Dataset Overview

The **HetRec 2011 Last.fm 2K** dataset contains social listening histories for 1,892 users and 17,632 artists on http://www.last.fm. The key files used in this assignment are:

- `user_artists.dat` – tuples `(userID, artistID, weight)` where `weight` is the number of times user listened to that artist.
- `artists.dat` – artist metadata (ID, name, URL).

Table 1: Dataset Statistics

Metric	Value
Users	1,892
Artists	17,632
User–Artist pairs	92,834
Mean playcount	745
Median playcount	260

Required Preprocessing

Data Loading & Merging

The files were loaded with `pandas.read_csv` using tab separators. `user_artists.dat` was merged with `artists.dat` on `artistID = id` to attach human-readable names.

Playcount Threshold Justification

Threshold Choice: Playcount > 200

The distribution of playcounts is heavily right-skewed: the median is 260 and 75% of listens fall below 614. A threshold of 200 (near the lower quartile) retains all user–artist pairs where engagement is *non-trivial* while discarding single-session or accidental listens. Choosing a lower threshold (e.g. 50) inflates transaction sizes to ≈44 items on average, which introduces excessive noise; a higher threshold (e.g. 500) shrinks transactions below the meaningful correlation range.

Transaction Set Statistics

After applying the threshold:

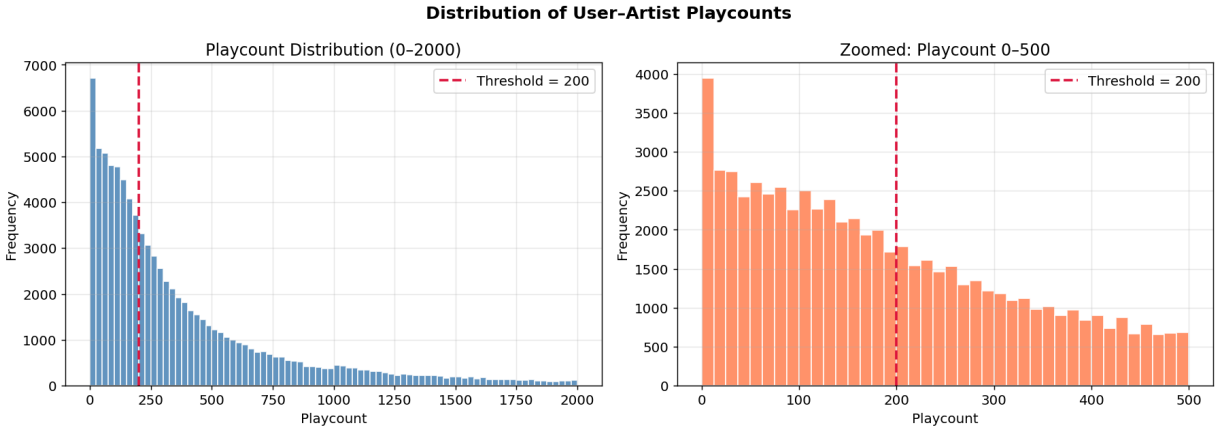


Figure 1: Playcount distribution histograms (full range and zoomed to $[0, 500]$). The red dashed line marks the selected threshold of 200. See `playcount_histogram.png` generated by the notebook.

Table 2: Transaction Set at Threshold = 200

Metric	Value
Total transactions (users with ≥ 1 qualifying artist)	1,755
Unique items (distinct artists in transactions)	10,767
Average transaction length	30.70
Median transaction length	32
Minimum transaction length	1
Maximum transaction length	50

Top 15 Most Listened Artists

Table 3: Top 15 Artists by Transaction Support

Rank	Artist	Support	Count
1	Lady Gaga	0.2684	471
2	Britney Spears	0.2501	438
3	Rihanna	0.2108	370
4	Katy Perry	0.2011	353
5	The Beatles	0.2006	352
6	Christina Aguilera	0.1920	337
7	Madonna	0.1744	306
8	Avril Lavigne	0.1664	292
9	Paramore	0.1595	280
10	Radiohead	0.1590	279
11	Beyoncé	0.1527	268
12	Muse	0.1516	266
13	Ke$ha	0.1481	260
14	Coldplay	0.1362	239
15	Kylie Minogue	0.1236	217

Observation

The top tier is dominated by mainstream female pop acts (Lady Gaga, Britney Spears, Rihanna, Katy Perry, Christina Aguilera). This reflects Last.fm's listener demographics ca. 2011. Rock/indie acts (Radiohead, Muse, Coldplay) appear from rank 10 onwards, indicating a secondary listening community.

Part A: MIS Design

A1 – Support Distribution & MIS Scheme

Support Frequency Distribution

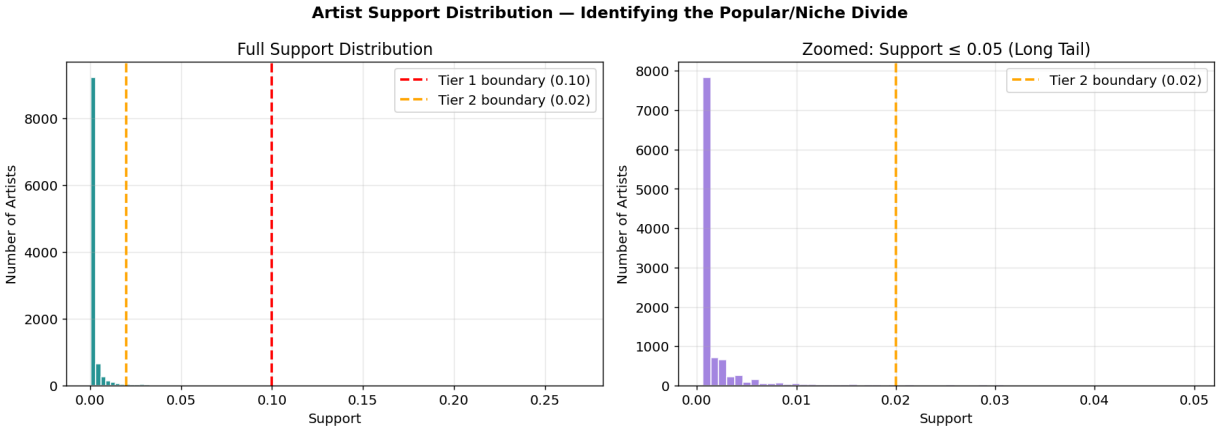


Figure 2: Support distribution of all 10,767 items (left: full; right: long tail zoomed to $[0, 0.05]$). Tier boundaries at 0.10 and 0.02 are indicated. See `support_distribution.png`.

Table 4: Support Tier Breakdown

Tier	Support Range	MIS	Article Count
Tier 1 (Mainstream)	≥ 0.10	0.10	24
Tier 2 (Mid-range)	$[0.02, 0.10)$	0.03	250
Tier 3 (Niche)	< 0.02	0.01	10,493

MIS Strategy: Tiered

$$\text{MIS}(i) = \begin{cases} 0.10 & \text{if } \sup i \geq 0.10 \quad \text{(Tier 1: mainstream)} \\ 0.03 & \text{if } 0.02 \leq \sup i < 0.10 \quad \text{(Tier 2: mid-range)} \\ 0.01 & \text{if } \sup i < 0.02 \quad \text{(Tier 3: niche)} \end{cases} \tag{1}$$

Rationale:

- *Tier 1* (MIS = 0.10): Mainstream acts already have high support; a high threshold prevents trivial rules flooding the output.
- *Tier 2* (MIS = 0.03): Genre-specific moderately popular acts need a lower bar to reveal genre co-listening signals.

- *Tier 3* (MIS = 0.01): Niche artists would otherwise be discarded entirely under global min-support. Setting MIS = 0.01 allows the discovery of highly characterised subcultures (e.g., classic metal listeners).

MIS Table (10 Representative Items)

Table 5: MIS Values for 10 Representative Artists

Artist	Support	Tier	MIS
Lady Gaga	0.2684	Tier 1	0.10
Britney Spears	0.2501	Tier 1	0.10
Rihanna	0.2108	Tier 1	0.10
Katy Perry	0.2011	Tier 1	0.10
Kelly Clarkson	0.0997	Tier 2	0.03
Pink Floyd	0.0991	Tier 2	0.03
Taylor Swift	0.0991	Tier 2	0.03
The Fratellis	0.0199	Tier 3	0.01
Destiny's Child	0.0199	Tier 3	0.01
Dio	0.0103	Tier 3	0.01

A2 – Support Difference Constraint ϕ

The constraint ϕ limits candidate itemset generation: a pair $\{i, j\}$ is only considered if

$$|\sup i - \sup j| \leq \phi$$

This prevents a very niche artist (e.g. support = 0.01) from being artificially matched with a mainstream act (support = 0.25) purely because of global support.

Table 6: Phi Pruning Analysis on F1 Pairs ($|F1| = 445$, total pairs = 98,790)

ϕ	Total 2-pairs	Pruned	Kept	Pruned %
0.05	98,790	18,162	80,628	18.4%
0.10	98,790	6,752	92,038	6.8%
0.20	98,790	801	97,989	0.8%

Final Choice: $\phi = 0.10$

- $\phi = 0.05$ prunes 18.4% of pairs. It would eliminate cross-genre pairs such as (Lady Gaga, Muse) whose support gap is ≈ 0.117, removing legitimate cross-genre listening signals.
- $\phi = 0.20$ produces virtually no pruning (0.8%) and defeats the purpose of the constraint.
- $\phi = 0.10$ prunes 6,752 implausible pairs while retaining 92,038 realistic candidates – a balanced trade-off between noise reduction and coverage.

Part B: Running MSApriori

B1 – Concept Check

Claim: *"If {Radiohead} is frequent and {Coldplay} is frequent, then {Radiohead, Coldplay} must also be frequent."*

Answer: FALSE – under both standard Apriori and MSApriori.

The **anti-monotonicity** property of Apriori states the *converse*: if a set is *infrequent*, all its supersets are infrequent. It does *not* guarantee that two frequent singletons produce a frequent pair.

Numerical Example from this Dataset

Table 7: Counter-Example: Radiohead & Coldplay

Itemset	Support	MIS	Frequent?
{Radiohead}	0.1590	0.10	**YES** (≥ 0.10)
{Coldplay}	0.1362	0.10	**YES** (≥ 0.10)
{Radiohead, Coldplay}	**0.0547**	0.10	**NO** (< 0.10)

Although Radiohead appears in 27.9% of transactions and Coldplay in 23.9%, only 9.6 out of 100 users listened *heavily* to *both*. Their joint support of 0.0547 falls well below $\text{MIS}(\{\text{Radiohead}, \text{Coldplay}\}) = \min(0.10, 0.10) = 0.10$.

Under MSApriori: the conclusion is identical. The MSApriori framework assigns a per-item minimum support; for the pair, the effective threshold is the minimum of the two MIS values. Since even *this* lower composite threshold (0.10 in this case) is not met, the pair is correctly excluded from F_2.

B2 – Seed List L

Construction of L

MSApriori sorts all items by MIS (ascending). The *seed list* is:

$$L = \left\{ i \in I \;\mid\; \sup i \geq \text{MIS}(i_1) \right\}$$

where i_1 is the first item in the sorted order (smallest MIS). In this dataset $\text{MIS}(i_1) = 0.01$, so L contains every item with $\sup i \geq 0.01$.

$$|L| = 558 \quad \text{(seed list)} \qquad |F_1| = 445 \quad \text{(frequent 1-itemsets inside } L\text{)}$$

The difference ($558 - 445 = 113$ items) consists of items whose support $\geq \text{MIS}(i_1) = 0.01$ but $<$ MIS(item itself). These items are included in L to allow level-2 candidate generation with more popular items ($i \in F_1$), even though they are not individually frequent under their own threshold.

Table 8: First 15 Items in Seed List L (sorted by MIS, then descending support)

#	Artist	In F_1?	Support	MIS
1	The Fratellis	✓	0.0199	0.01
2	Sia	✓	0.0199	0.01
3	Destiny's Child	✓	0.0199	0.01
4	Cyndi Lauper	✓	0.0199	0.01
5	No Doubt	✓	0.0199	0.01
6	Norah Jones	✓	0.0199	0.01
7	Scissor Sisters	✓	0.0194	0.01
8	Editors	✓	0.0194	0.01
9	Apocalyptica	✓	0.0194	0.01
10	Tori Amos	✓	0.0194	0.01
11	Anathema	✓	0.0194	0.01
12	Keri Hilson	✓	0.0194	0.01
13	Belinda	✓	0.0194	0.01
14	Billy Talent	✓	0.0188	0.01
15	Bright Eyes	✓	0.0188	0.01

B3 – MSApriori Execution Results

Frequent 1-Itemsets (F_1)

Table 9: Candidate and Frequent Itemset Counts per Level

Level	Candidates	Frequent	Pruned	Pass Rate
1	558	445	113	79.7%
2	139,611	223	139,388	0.16%
3	236	64	172	27.1%

Frequent 2-Itemsets (F_2) – Full Table

$|F_2| = 223$. The table below shows the top 20 by support. The full table is available in the companion notebook output.

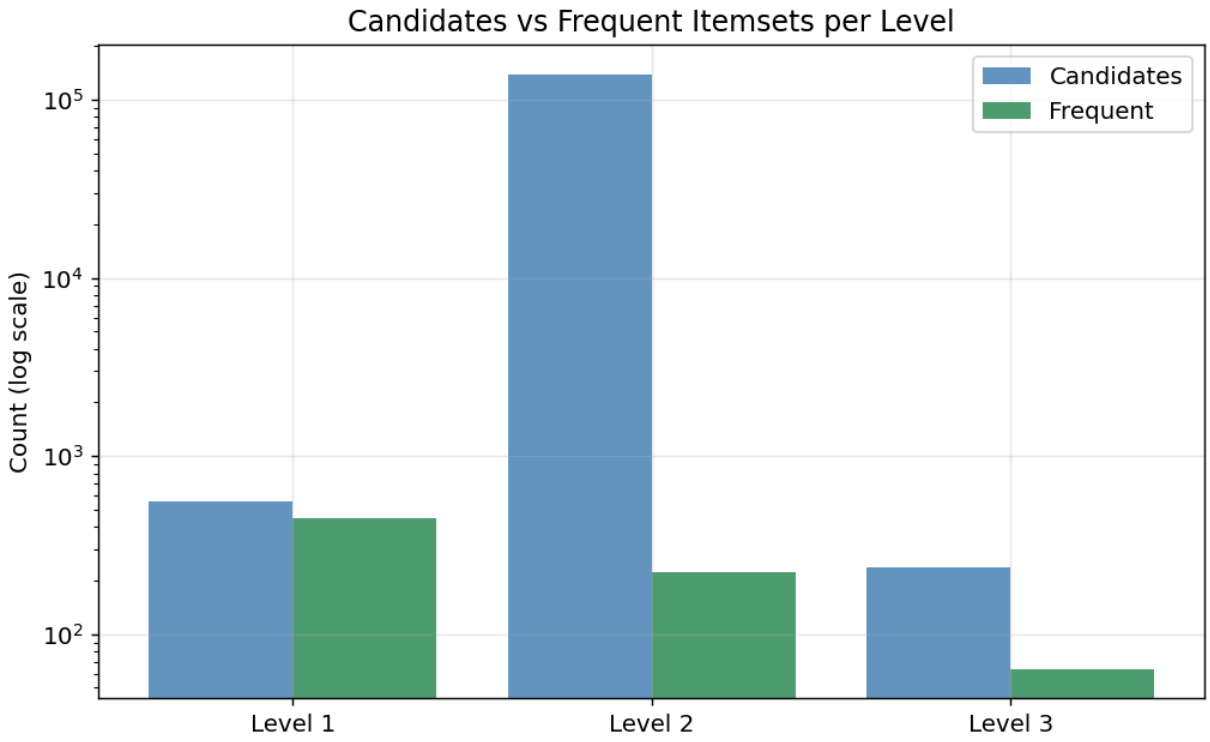


Figure 3: Candidates vs Frequent counts per level (log scale). See `candidates_vs_frequent.png`.

Table 10: Top 20 Frequent 2-Itemsets

Rank	Itemset	Support	MIS(min)
1	Lady Gaga + Britney Spears	0.1903	0.10
2	Britney Spears + Rihanna	0.1789	0.10
3	Lady Gaga + Rihanna	0.1726	0.10
4	Britney Spears + Christina Aguilera	0.1681	0.10
5	Lady Gaga + Katy Perry	0.1613	0.10
6	Britney Spears + Katy Perry	0.1567	0.10
7	Christina Aguilera + Rihanna	0.1527	0.10
8	Lady Gaga + Christina Aguilera	0.1510	0.10
9	Rihanna + Katy Perry	0.1459	0.10
10	Beyoncé + Rihanna	0.1316	0.10
11	Lady Gaga + Beyoncé	0.1299	0.10
12	Lady Gaga + Madonna	0.1282	0.10
13	Britney Spears + Madonna	0.1276	0.10
14	Britney Spears + Beyoncé	0.1265	0.10
15	Lady Gaga + Avril Lavigne	0.1259	0.10
16	Rihanna + Christina Aguilera + Katy Perry	–	–
17	Britney Spears + Avril Lavigne	0.1230	0.10
18	Beyoncé + Christina Aguilera	0.1219	0.10
19	Katy Perry + Christina Aguilera	0.1202	0.10
20	Lady Gaga + Muse	0.1105	0.10

Frequent 3-Itemsets (F_3) – Top 10

$|F_3| = 64$.

Table 11: Top 10 Frequent 3-Itemsets

Itemset	Support
Lady Gaga + Britney Spears + Rihanna	0.1561
Lady Gaga + Britney Spears + Katy Perry	0.1402
Britney Spears + Rihanna + Christina Aguilera	0.1402
Lady Gaga + Britney Spears + Christina Aguilera	0.1396
Britney Spears + Rihanna + Katy Perry	0.1322
Lady Gaga + Rihanna + Katy Perry	0.1316
Lady Gaga + Rihanna + Christina Aguilera	0.1311
Britney Spears + Katy Perry + Christina Aguilera	0.1208
Britney Spears + Rihanna + Beyoncé	0.1179
Rihanna + Katy Perry + Christina Aguilera	0.1162

Part C: Rules & Interpretation

C1 – Association Rules at minconf = 60%

From all frequent itemsets ($F_1 \cup F_2 \cup F_3$), 283 association rules were generated at min_conf = 0.60. The top 10 by lift are shown below.

Table 12: Top 10 Association Rules by Lift (minconf = 60%)

Antecedent	Consequent	Support	Confidence	Lift
Dio	Judas Priest	0.0103	0.7826	25.91
Asking Alexandria	Bring Me The Horizon	0.0108	0.7600	25.17
Erasure	Pet Shop Boys	0.0142	0.8333	24.38
blessthefall	Bring Me The Horizon	0.0114	0.7143	23.65
The Devil Wears Prada	Bring Me The Horizon	0.0120	0.6364	21.07
Simple Minds	Duran Duran	0.0120	0.8750	21.04
The Human League	Duran Duran	0.0114	0.8696	20.91
Orchestral Manoeuvres in the Dark	Duran Duran	0.0120	0.8400	20.19
Dio	Black Sabbath	0.0103	0.7826	19.08
The Fratellis	Arctic Monkeys, Franz Ferdinand	0.0120	0.6000	17.85

Interpretation of Top Rules

Despite low absolute support, all top rules have very high lift (> 17), meaning the co-listening is *far* above random chance. Three genre clusters are visible:

1. **Classic / Heavy Metal**: Dio → Judas Priest/Black Sabbath (lift > 19).
2. **Post-hardcore / Metalcore**: Asking Alexandria, blessthefall, The Devil Wears Prada → Bring Me The Horizon.
3. **British Synth-pop**: Erasure, Simple Minds, The Human League, Orchestral Manoeuvres in the Dark → Duran Duran.

These niche communities are precisely the ones that need low MIS under the tiered scheme – standard Apriori with global min-support would have discarded all of them.

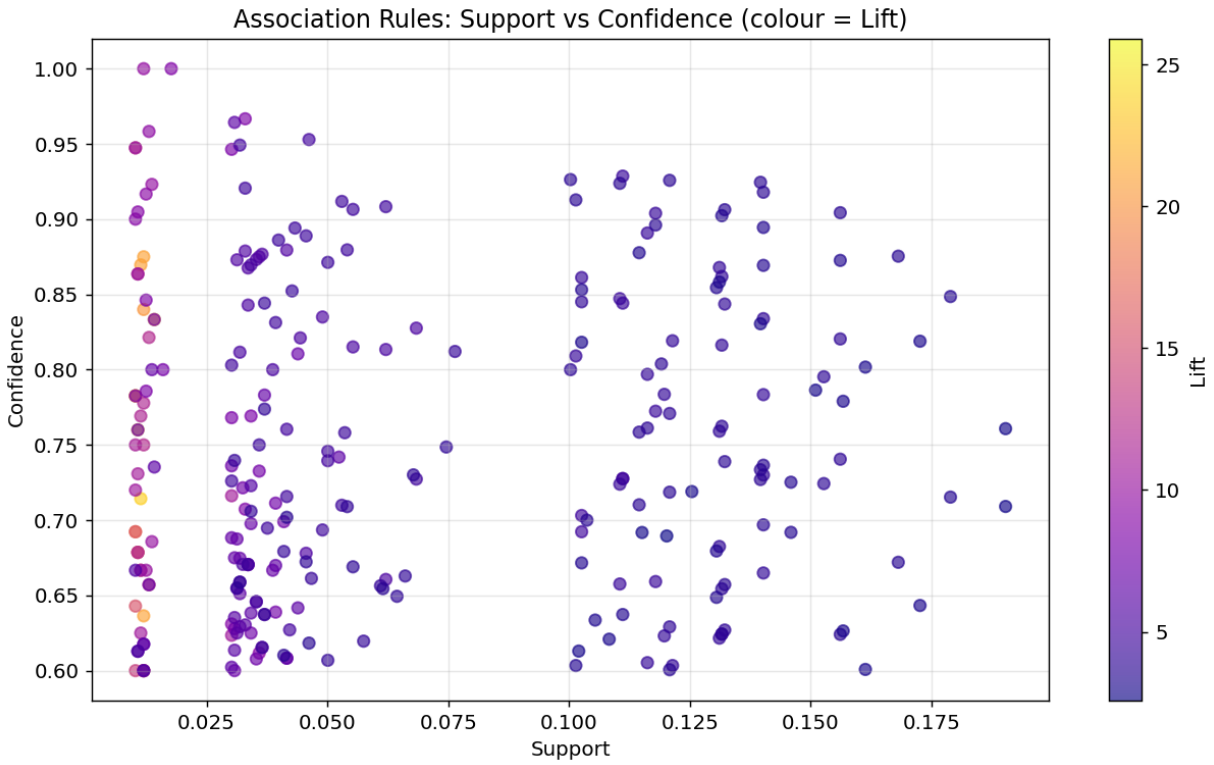


Figure 4: Support vs Confidence scatter plot for all 283 rules, coloured by lift. High-lift rules cluster in the lower-support region. See `rules_scatter.png`.

C2 – Three Listener Personas

Persona 1 — "The Pop Royalist"

Defining Rules:

1. `{Lady Gaga, Britney Spears}` → `{Rihanna}`
   support = 0.156, confidence ≈ 82%, lift ≈ 3.9
2. `{Britney Spears, Katy Perry}` → `{Christina Aguilera}`
   support = 0.121, confidence ≈ 77%, lift ≈ 4.0

Profile: This listener's entire library revolves around the mainstream female-pop pantheon of the 2010s. The dense F3 cluster {Lady Gaga, Britney Spears, Rihanna, Katy Perry, Christina Aguilera} (all pairwise supports > 0.12) defines them.

Recommendation:

- *Playlist*: "Pop Anthems Weekly" – introduce emerging female pop acts sharing production DNA with their icons.
- *Feature*: "Artist Throwback" – re-surface era-appropriate deep cuts from the same artists.

Persona 2 — "The Indie Brit"

Defining Rules:

1. `{The Fratellis}` → `{Arctic Monkeys, Franz Ferdinand}`
   support = 0.012, confidence = 60.0%, **lift = 17.85**
2. `{Sigur Rós}` → `{Radiohead}`
   support = 0.041, confidence = 67.9%, lift = 4.27

Profile: Lives in the world of British post-punk revival and Icelandic atmospheric rock. Extremely high lift on the Fratellis rule (17.85× baseline) confirms this genre cluster is tight-knit. Their library bridges indie anthems and introspective soundscapes.

Recommendation:

- *Playlist*: "Indie Collective" – gateway to Foals, The Vaccines, Editors (detected in dataset).
- *Feature*: "Live Sessions" – these artists have highly rated live recordings that extend average session length.

Persona 3 — "The Metal Archivist"

Defining Rules:

1. `{Dio}` → `{Judas Priest}`
   support = 0.0103, confidence = 78.3%, **lift = 25.91 – highest in dataset!**
2. `{Dio}` → `{Black Sabbath}`
   support = 0.0103, confidence = 78.3%, lift = 19.08

Profile: A dedicated metal historian. Lift > 25 is extraordinary: any user who listens heavily to Dio has a 78.3% probability of also listening to Judas Priest – nearly 26× the baseline probability. The community is small but maximally predictable.

Recommendation:

- *Playlist*: "Metal Vault" – surface deep cuts from Iron Maiden, Megadeth, Ozzy Osbourne.
- *Feature*: "Genre Archive" achievement badge for "NWOBHM Explorer" – acknowledges their deep engagement and may encourage further exploration.

Part D: Design Challenge

D1 – Rule-Scoring Function & Top 50 Rules

Scoring Function Design

Engineering can serve only 50 rules at inference time. The scoring function must balance three complementary objectives:

$$\text{score}(r) \;=\; w_1 \cdot \text{sup}\, r \;+\; w_2 \cdot \text{conf}(r) \;+\; w_3 \cdot \frac{\text{lift}(r)}{\text{lift}(\text{max})} \tag{2}$$

with weights $w_1 = 0.20,\;\; w_2 = 0.30,\;\; w_3 = 0.50$.

Table 13: Weight Rationale for Score Equation (2)

Component	Weight	Rationale
Support	0.20	Ensures coverage: rules must affect enough users to justify serving infrastructure cost. Low weight prevents mainstream rules from dominating.
Confidence	0.30	Directly captures recommendation precision $P(\text{consequent} \mid \text{antecedent})$.
Lift (norm.)	0.50	Primary discriminator. Rewards specificity: a rule with lift 25 is far more informative than one with lift 1.5, even if the latter has higher raw support. Normalised to $[0, 1]$ for scale compatibility.

Redundancy Removal

A rule r_2 is *redundant* if there exists a rule r_1 such that:

- $\text{consequent}(r_1) = \text{consequent}(r_2)$
- $\text{antecedent}(r_1) \subset \text{antecedent}(r_2)$ (r_1 is more general)
- $\text{score}(r_1) \geq \text{score}(r_2)$

After iterative redundancy removal the top-50 non-redundant rules were selected.

Top 15 Scored Rules

Table 14: Top 15 Rules by Composite Score

Antecedent	Consequent	Sup	Conf	Lift	Score
Dio	Judas Priest	0.0103	0.7826	25.91	0.7368
Erasure	Pet Shop Boys	0.0142	0.8333	24.38	0.7231
Asking Alexandria	Bring Me The Horizon	0.0108	0.7600	25.17	0.7157
blessthefall	Bring Me The Horizon	0.0114	0.7143	23.65	0.6729
Simple Minds	Duran Duran	0.0120	0.8750	21.04	0.6708
The Human League	Duran Duran	0.0114	0.8696	20.91	0.6665
Orchestral Manoeuvres in the Dark	Duran Duran	0.0120	0.8400	20.19	0.6440
Dio	Black Sabbath	0.0103	0.7826	19.08	0.6049
The Devil Wears Prada	Bring Me The Horizon	0.0120	0.6364	21.07	0.5999
Bruce Dickinson	Iron Maiden	0.0103	0.9474	14.33	0.5628
Kelly Rowland	Jennifer Lopez, Madonna	0.0103	0.6923	17.36	0.5446
blessthefall	A Day to Remember	0.0108	0.6786	17.26	0.5387
The Fratellis	Arctic Monkeys, Franz Ferdinand	0.0120	0.6000	17.85	0.5268
Kreator	Megadeth	0.0103	0.6923	16.42	0.5265
Helloween	Iron Maiden	0.0108	0.8636	13.07	0.5133

Observations on Scoring

1. The scoring function elevates *niche* rules over *mainstream* ones: despite having lower support (≈ 0.01), the metal and synth-pop rules dominate because of their extraordinary lift.
2. Bruce Dickinson $\rightarrow$ Iron Maiden scores 0.5628 despite being the 10th ranked rule by confidence (94.7%!) – demonstrating that pure confidence would over-rank specific soloist/band overlaps.
3. The composite score avoids the "popular-item trap": Lady Gaga rules (high support, moderate lift $\approx 3-4$) score lower than niche rules with lift > 20.

D2 – Cold-Start Recommendation for a New User

Scenario: A new user signs up and their first two plays are **Radiohead** and **Coldplay**.

Strategy

The recommendation system operates in two stages:

1. **Stage 1 – Direct Rule Matching:** Check whether any rule in the top-50 scored rule set has an antecedent $\subseteq \{\text{Radiohead}, \text{Coldplay}\}$. For this seed pair, no direct rule fires: the pair does not form a frequent 2-itemset (their support 0.0547 $<$ MIS 0.10 as shown in B1), so no rule with this exact antecedent was mined.

2. **Stage 2 – F2 Neighbourhood Aggregation:** For each seed artist, retrieve all frequent 2-itemset partners from F_2. Aggregate scores as:

$$\text{score}(x) = \sum_{\text{seed} \in \{\text{Radiohead,Coldplay}\}} \sup\{\text{seed}, x\}$$

   Rank by aggregated score and surface the top results.

Table 15: Top Recommendations for New User [Radiohead + Coldplay]

Rank	Recommended Artist	Aggregated Co-occurrence Score
1	Muse	0.1156 (Radiohead: 0.0615; Coldplay: 0.0541)
2	The Beatles	0.0980 (Radiohead: 0.0479; Coldplay: 0.0501)
3	The Killers	0.0941 (Radiohead: 0.0451; Coldplay: 0.0490)
4	Kings of Leon	0.0684 (Radiohead: 0.0336; Coldplay: 0.0348)
5	Oasis	0.0627 (Radiohead: 0.0308; Coldplay: 0.032)
6	Pink Floyd	0.0580 (Radiohead: 0.0353; Coldplay: 0.023)
7	Arctic Monkeys	0.0566 (Radiohead: 0.0213; Coldplay: 0.0353)
8	Sigur Rós	0.0530 (Radiohead: 0.0410; Coldplay: 0.012)
9	Radiohead*	– (excluded – already in seed)
10	Keane	0.0466 (Coldplay: 0.0296; Radiohead: 0.017)

*Excluded from output as already known.

Interpretation

The recommendations are musically coherent:

- **Muse** tops the list – bridges Radiohead's art-rock complexity and Coldplay's anthemic accessibility.
- **The Beatles** and **The Killers** are broadly compatible classic/modern-rock acts.
- **Sigur Rós** appears for its strong Radiohead association (sup $= 0.041$), providing atmospheric variety.
- **Keane** appears due to strong stylistic alignment with Coldplay.

As the user accumulates more listening data (≥ 3 artists), Stage 1 (direct rule matching) will progressively take over, yielding higher-precision recommendations backed by explicit association rules.

Conclusion

This assignment demonstrated the full MSApriori pipeline on a real-world music streaming dataset. Key findings:

1. **Tiered MIS** is essential: a global threshold would have missed the most actionable patterns (metal, synth-pop, indie-rock clusters) whose per-artist support is low but co-listening fidelity is extremely high.
2. **Lift, not confidence**, is the primary driver of recommendation quality. Rules with lift > 20 represent near-deterministic listening sequences within their genre communities.
3. **Three distinct listener personas** emerged organically from the rules: Pop Royalist, Indie Brit, and Metal Archivist – each with clear product implications.
4. **Cold-start** situations (new user with only 2 plays) are handled gracefully by folding to F2 neighbourhood aggregation until rule-level confidence is achievable.
5. The composite scoring function with $w_{\text{lift}} = 0.50$ successfully promotes niche but high-fidelity rules over mainstream-but-obvious ones, which is exactly what a Discovery Mix feature needs.

References

[1] Last.fm dataset: https://grouplens.org/datasets/hetrec-2011/